Shower Shape Comparison: Nominal vs Double-Interaction Mixed MC

0 mrad (18.7% DI) and 1.5 mrad (7.2% DI) Crossing Angles

PPG12 Analysis Team (automated report)

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1 Introduction

In *pp* collisions at RHIC, two independent collisions can occur in the same bunch crossing (double interaction / pileup), distorting reconstructed cluster kinematics and shower shapes. To quantify this effect, full GEANT4 double-interaction MC samples (`photon10_double`, `jet12_double`) are blended with nominal single-interaction MC at the physics-dictated fractions:

- **0 mrad** crossing angle (runs 47289–51274, $\mathcal{L} = 32.7$ pb^{-1}): 18.7% double interaction
- **1.5 mrad** crossing angle (runs 51274–54000, $\mathcal{L} = 16.9$ pb^{-1}): 7.2% double interaction

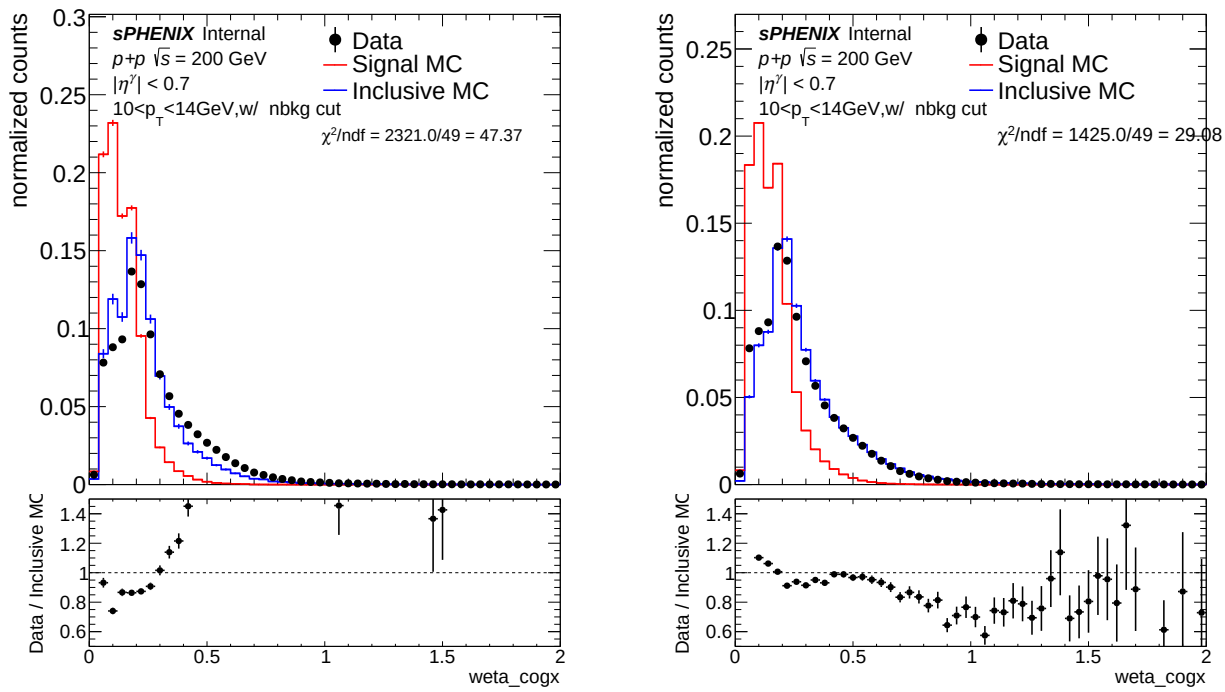
The blending uses a two-pass pipeline (`run_showershape_double.sh`): Pass 1 builds a combined vertex distribution (weighted `hadd` of single and double vtxscan files); Pass 2 reruns the full shower shape analysis with consistent vertex reweighting and `mix_weight` applied to all histogram fills. The final "mixed MC" is the `hadd` of single-weighted and double-weighted output files.

All comparisons use the **cut1** selection level (with NPB preselection). Each plot shows data (black points), signal MC (red), and inclusive MC (blue), area-normalized, with a χ^2/ndf annotation and a Data/Inclusive MC ratio panel. Left panels show nominal (single-only) MC; right panels show mixed (double-blended) MC.

2 0 mrad Crossing Angle (18.7% Double Interaction)

2.1 w_η^{CoG} (Eta Cluster Width)

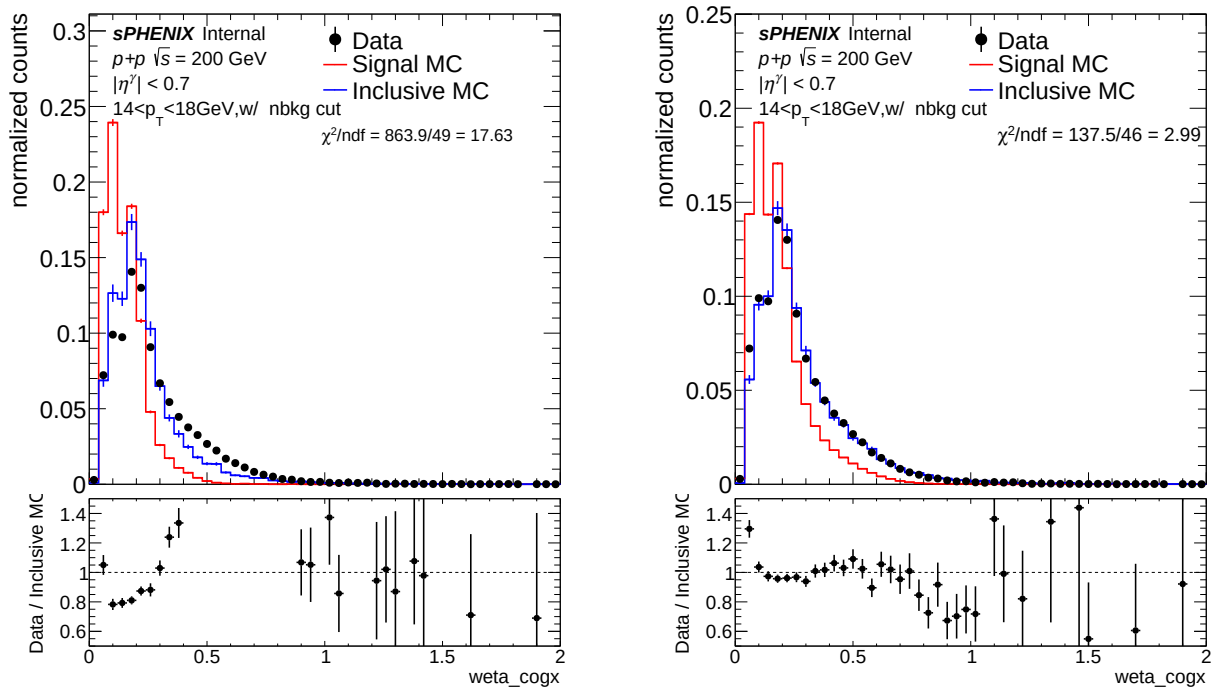
w_η^{CoG} is strongly affected by double interactions: vertex shifts displace clusters along the beam axis, broadening the η profile.



(a) Nominal MC

(b) Mixed MC (18.7% DI)

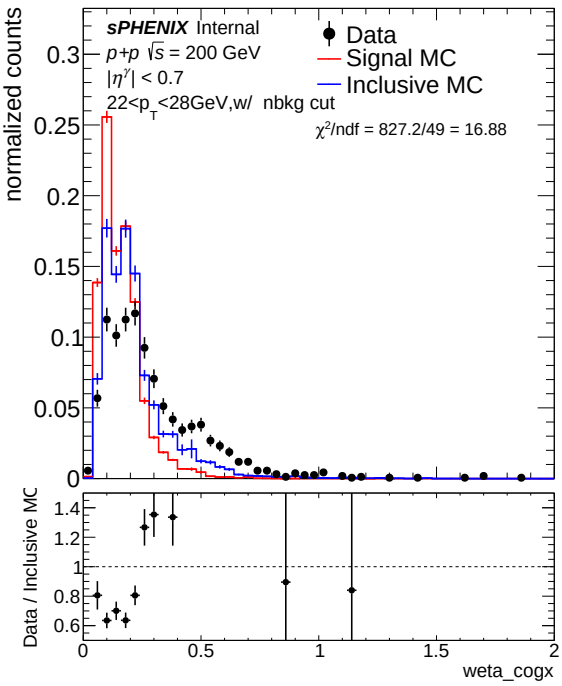
Figure 1: w_η^{CoG}, $10 < E_T < 14\,\mathrm{GeV}$ (pt0), $0\,\mathrm{mrad}$. Nominal $\chi^2/\mathrm{ndf} = 47.37$; mixed $\chi^2/\mathrm{ndf} = 29.08$ ($1.6\times$ improvement).



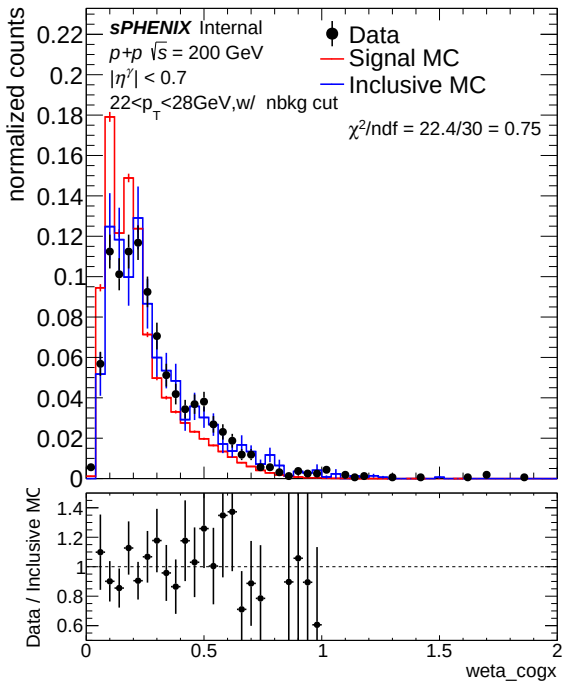
(a) Nominal MC

(b) Mixed MC (18.7% DI)

Figure 2: w_η^{CoG}, $14 < E_T < 18\,\mathrm{GeV}$ (pt1), $0\,\mathrm{mrad}$. Nominal $\chi^2/\mathrm{ndf} = 17.63$; mixed $\chi^2/\mathrm{ndf} = 2.99$ ($5.9\times$ improvement).



(a) Nominal MC



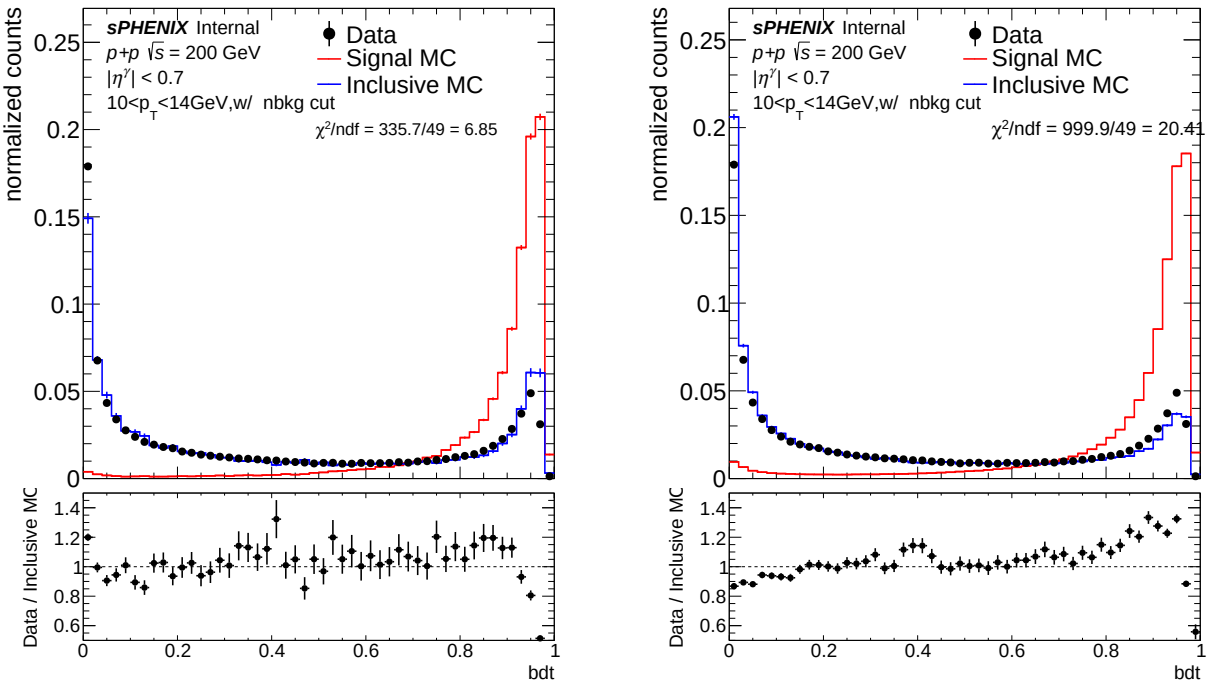
(b) Mixed MC (18.7% DI)

Figure 3: w_η^{CoG}, $22 < E_T < 28\,\mathrm{GeV}$ (pt3), 0 mrad. Nominal $\chi^2/\mathrm{ndf} = 16.88$; mixed $\chi^2/\mathrm{ndf} = 0.75$ (22.5× improvement).

The mixed MC dramatically improves data–MC agreement for w_η^{CoG} across all E_T bins at 0 mrad, with χ^2/ndf dropping by factors of 1.6 (pt0) to 22.5 (pt3).

2.2 BDT Score

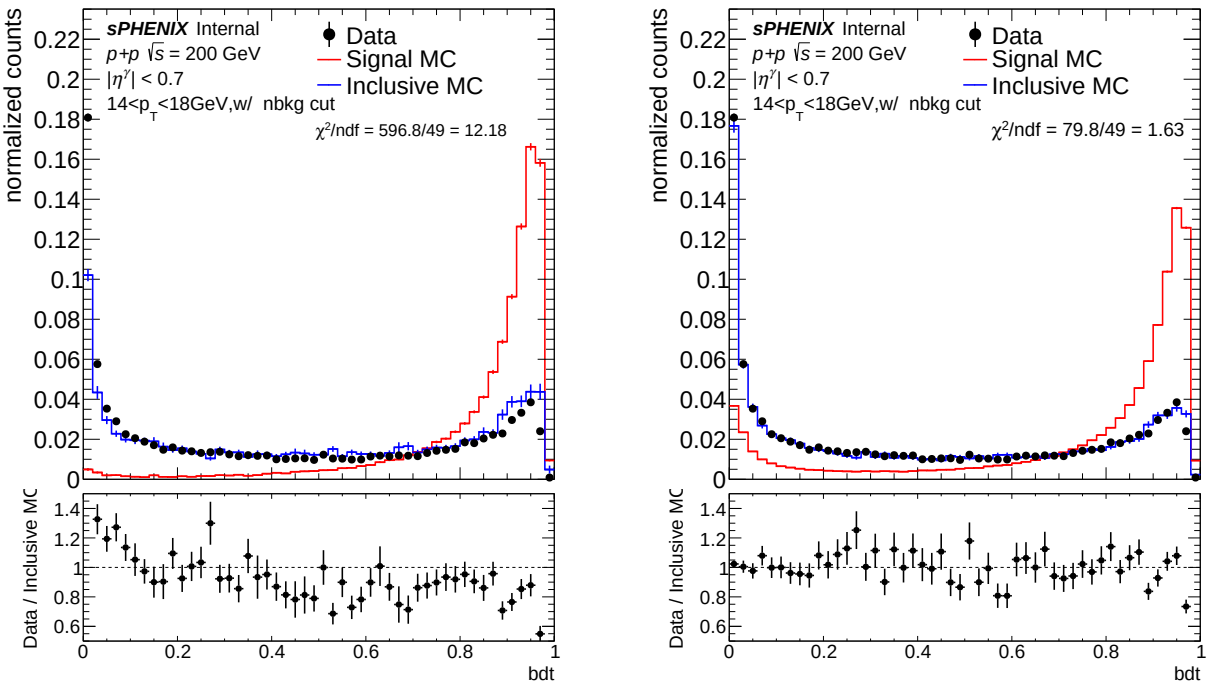
The photon/jet BDT discriminator amplifies underlying shower shape distortions through the XGBoost decision function.



(a) Nominal MC

(b) Mixed MC (18.7% DI)

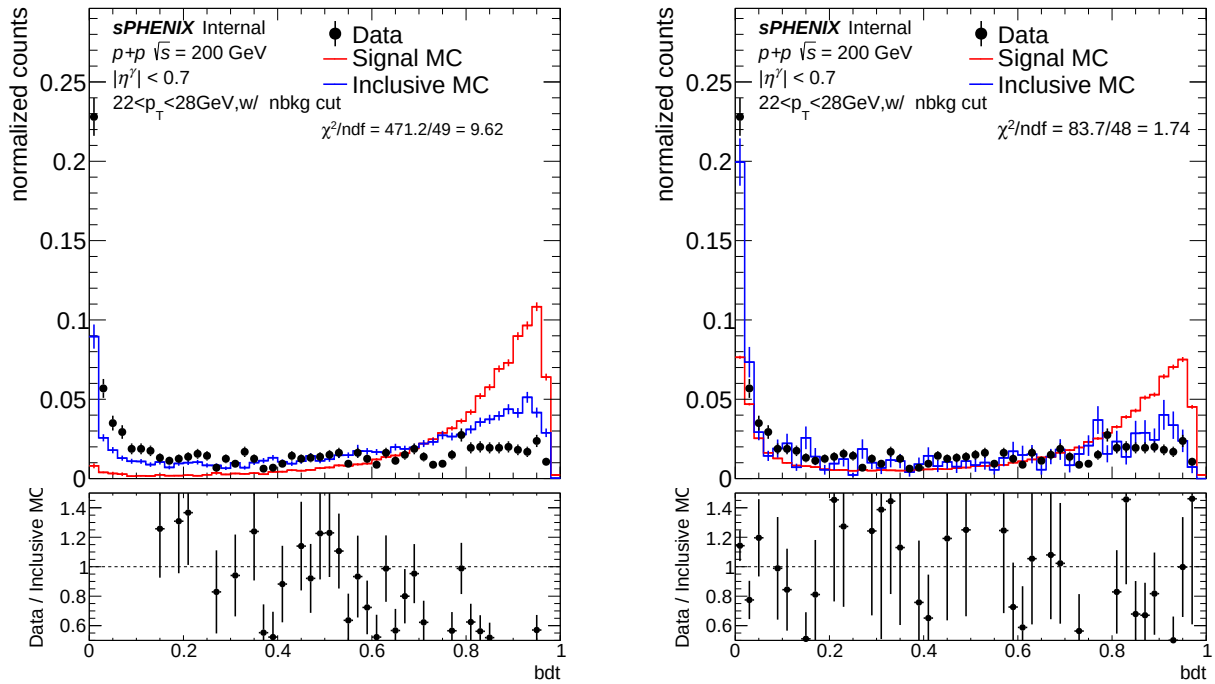
Figure 4: BDT score, $10 < E_T < 14\,\text{GeV}$ (pt0), 0 mrad. Nominal $\chi^2/\text{ndf} = 6.85$; mixed $\chi^2/\text{ndf} =$ 20.41 (3.0× **degradation**).



(a) Nominal MC

(b) Mixed MC (18.7% DI)

Figure 5: BDT score, $14 < E_T < 18\,\text{GeV}$ (pt1), 0 mrad. Nominal $\chi^2/\text{ndf} = 12.18$; mixed $\chi^2/\text{ndf} =$ 1.63 (7.5× improvement).



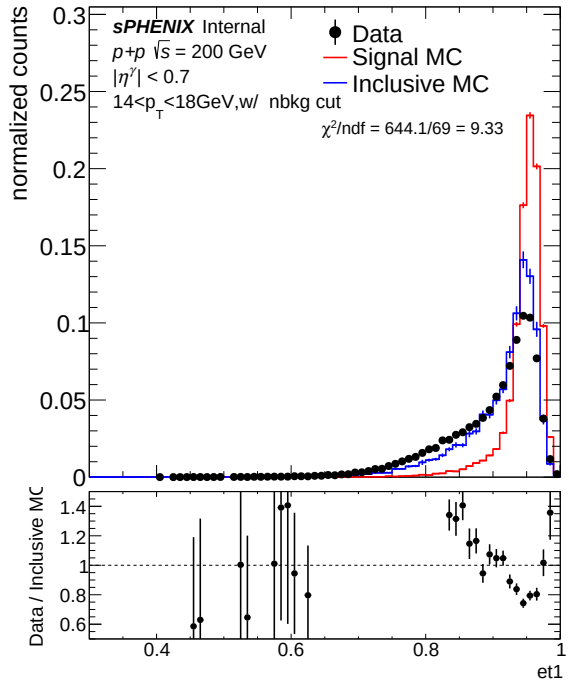
(a) Nominal MC

(b) Mixed MC (18.7% DI)

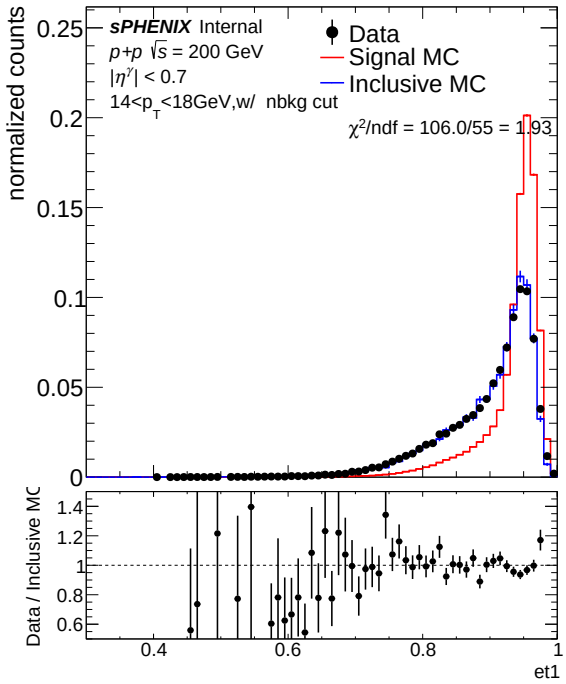
Figure 6: BDT score, $22 < E_T < 28\,\text{GeV}$ (pt3), 0 mrad. Nominal $\chi^2/\text{ndf} = 9.62$; mixed $\chi^2/\text{ndf} = 1.74$ (5.5× improvement).

At 0 mrad, the BDT χ^2/ndf improves substantially at pt1 (7.5×) and pt3 (5.5×). However, at pt0, the mixed MC *degrades* agreement by 3.0× (6.85 → 20.41). This low-E_T degradation was not reported in the original 0 mrad analysis and is now understood to be a common feature of both crossing angles (see Section 6).

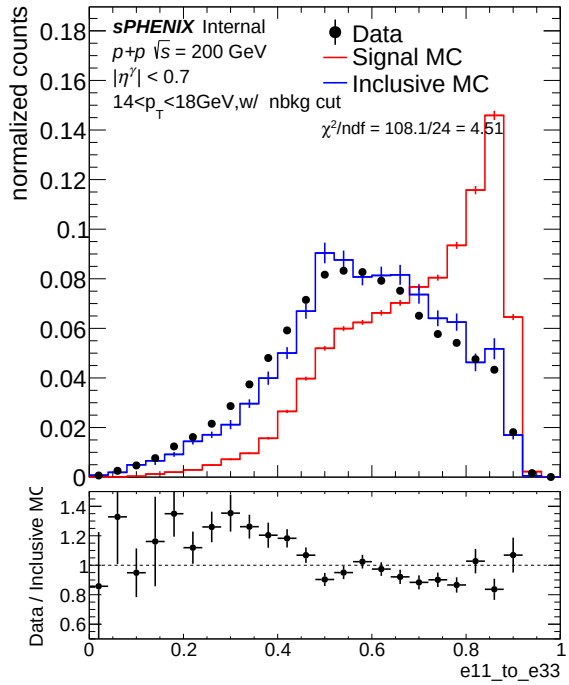
2.3 f_1 and $E_{1\times1}/E_{3\times3}$



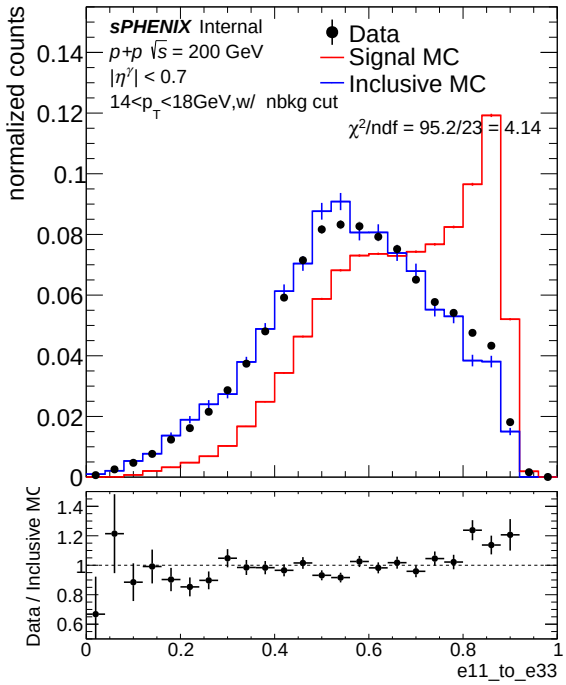
(a) f_1, Nominal MC



(b) f_1, Mixed MC (18.7% DI)



(c) $E_{1\times1}/E_{3\times3}$, Nominal MC

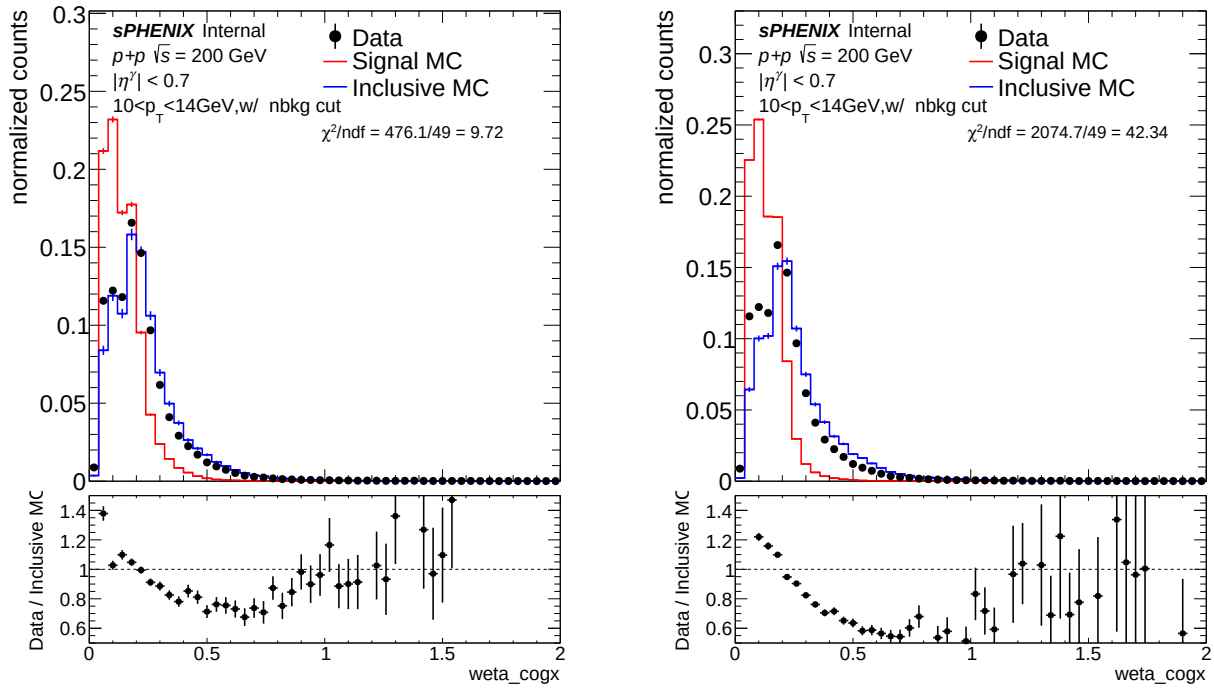


(d) $E_{1\times1}/E_{3\times3}$, Mixed MC (18.7% DI)

Figure 7: f_1 (top) and $E_{1\times1}/E_{3\times3}$ (bottom) at $14 < E_T < 18\,\text{GeV}$ (pt1), 0 mrad. f_1: $\chi^2/\text{ndf} = 9.33 \rightarrow 1.93$ (4.8×). $E_{1\times1}/E_{3\times3}$: $\chi^2/\text{ndf} = 4.51 \rightarrow 4.14$ (1.1×).

3 1.5 mrad Crossing Angle (7.2% Double Interaction)

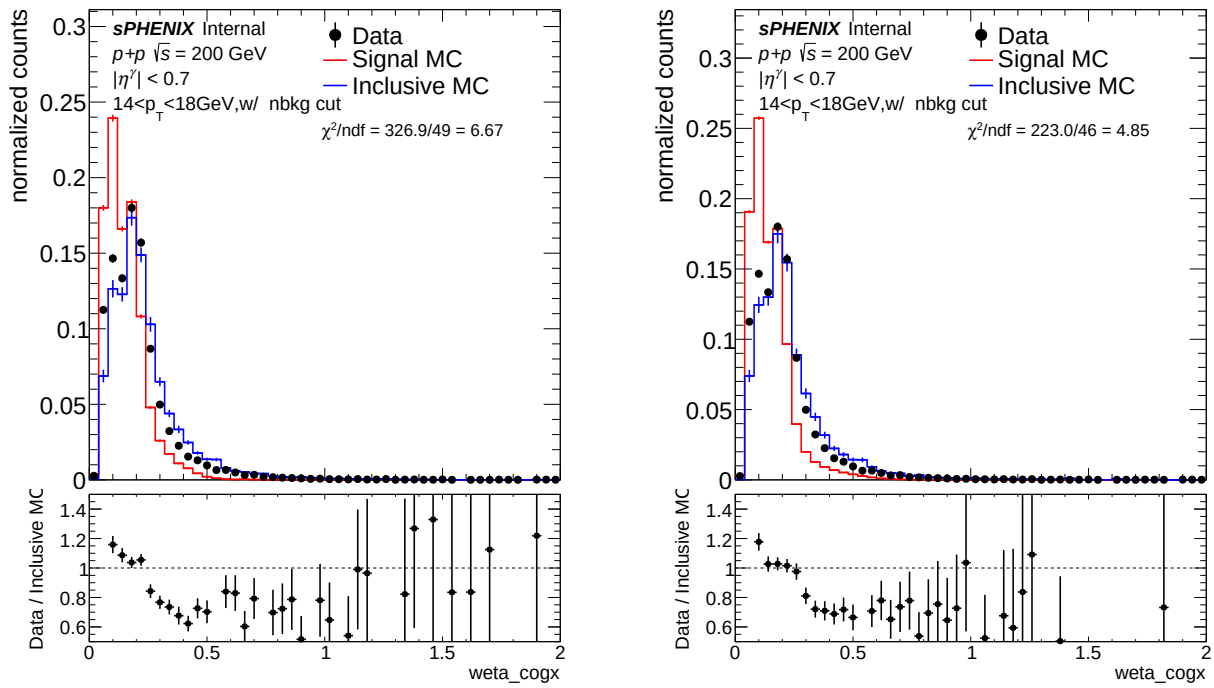
3.1 w_η^{CoG} (Eta Cluster Width)



(a) Nominal MC

(b) Mixed MC (7.2% DI)

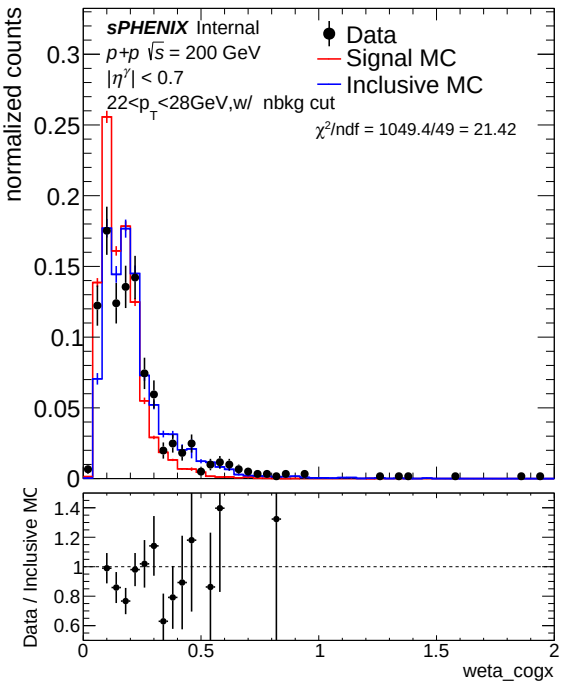
Figure 8: w_η^{CoG}, $10 < E_T < 14\,\mathrm{GeV}$ (pt0), $1.5\,\mathrm{mrad}$. Nominal $\chi^2/\mathrm{ndf} = 9.72$; mixed $\chi^2/\mathrm{ndf} = 42.34$ ($4.4\times$ **degradation**).



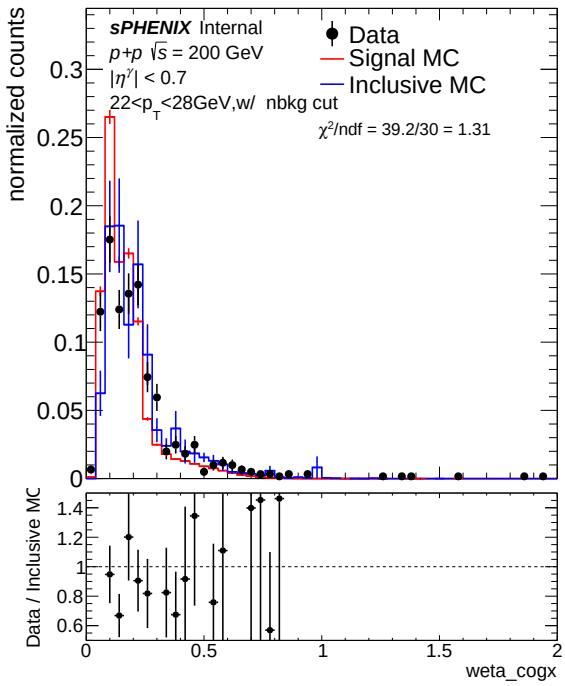
(a) Nominal MC

(b) Mixed MC (7.2% DI)

Figure 9: w_η^{CoG}, $14 < E_T < 18\,\mathrm{GeV}$ (pt1), $1.5\,\mathrm{mrad}$. Nominal $\chi^2/\mathrm{ndf} = 6.67$; mixed $\chi^2/\mathrm{ndf} = 4.85$ ($1.4\times$ improvement).



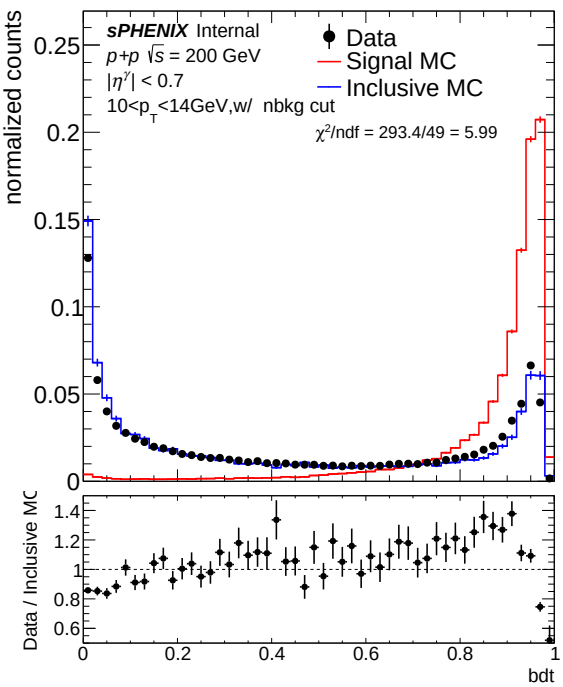
(a) Nominal MC



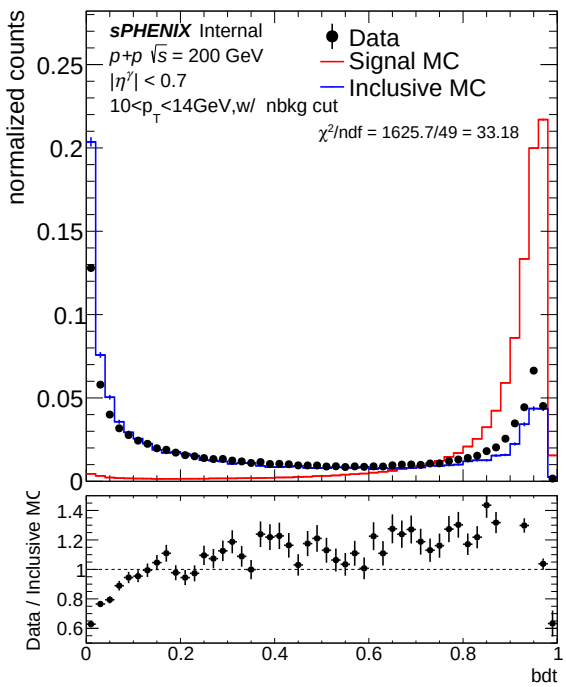
(b) Mixed MC (7.2% DI)

Figure 10: w_η^{CoG}, $22 < E_T < 28\,\mathrm{GeV}$ (pt3), 1.5 mrad. Nominal $\chi^2/\mathrm{ndf} = 21.42$; mixed $\chi^2/\mathrm{ndf} = 1.31$ (16.3× improvement).

3.2 BDT Score

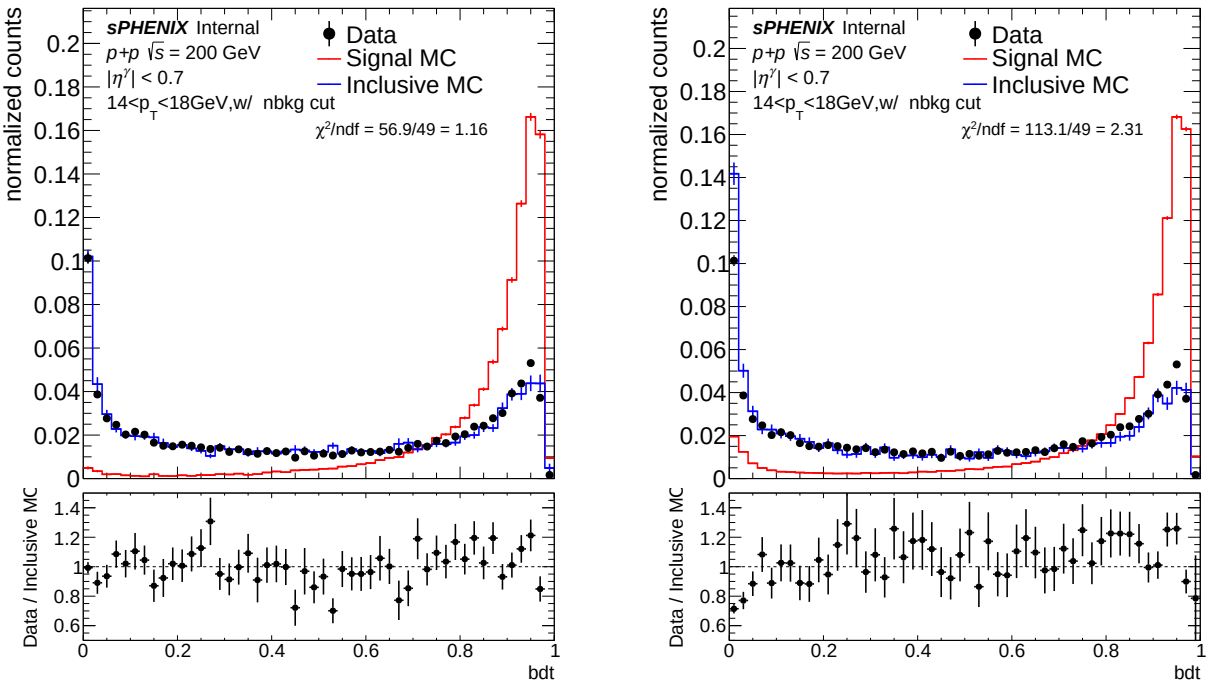


(a) Nominal MC



(b) Mixed MC (7.2% DI)

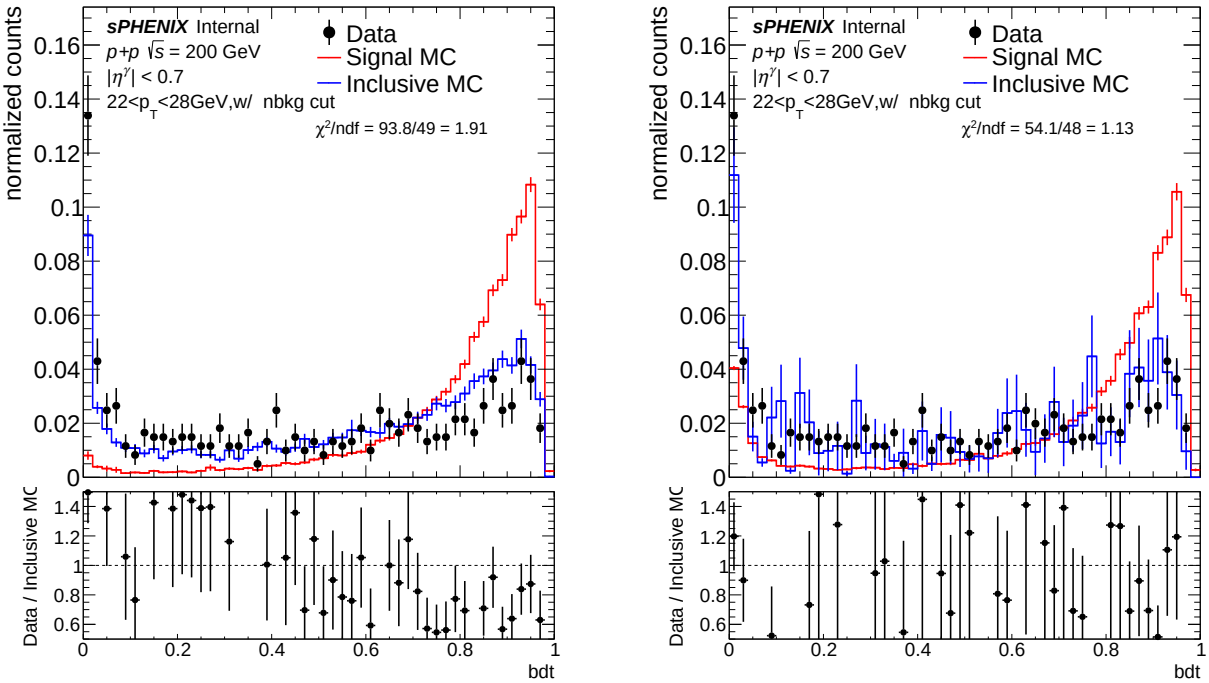
Figure 11: BDT score, $10 < E_T < 14\,\mathrm{GeV}$ (pt0), 1.5 mrad. Nominal $\chi^2/\mathrm{ndf} = 5.99$; mixed $\chi^2/\mathrm{ndf} = 33.18$ (5.5× **degradation**).



(a) Nominal MC

(b) Mixed MC (7.2% DI)

Figure 12: BDT score, $14 < E_T < 18\,\text{GeV}$ (pt1), 1.5 mrad. Nominal $\chi^2/\text{ndf} = 1.16$; mixed $\chi^2/\text{ndf} = 2.31$ (2.0× **degradation**; nominal already excellent).

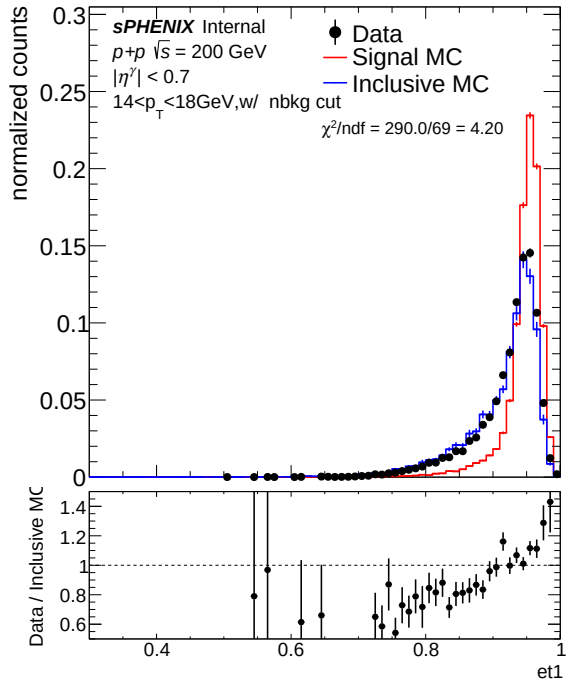


(a) Nominal MC

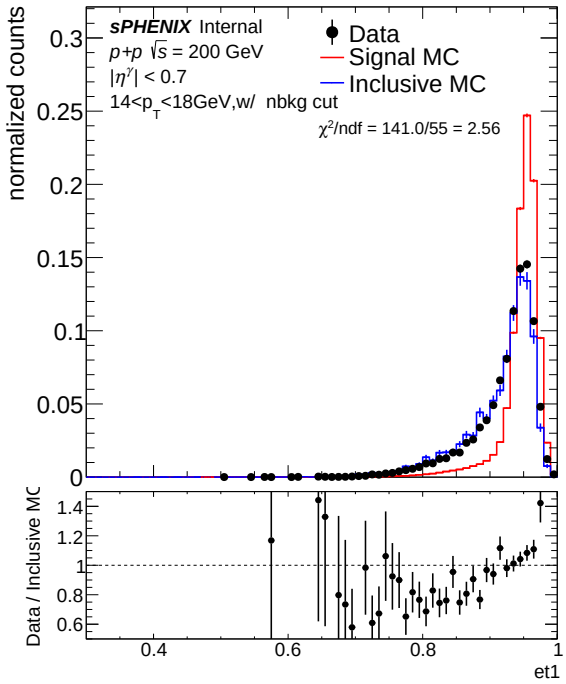
(b) Mixed MC (7.2% DI)

Figure 13: BDT score, $22 < E_T < 28\,\text{GeV}$ (pt3), 1.5 mrad. Nominal $\chi^2/\text{ndf} = 1.91$; mixed $\chi^2/\text{ndf} = 1.13$ (1.7× improvement).

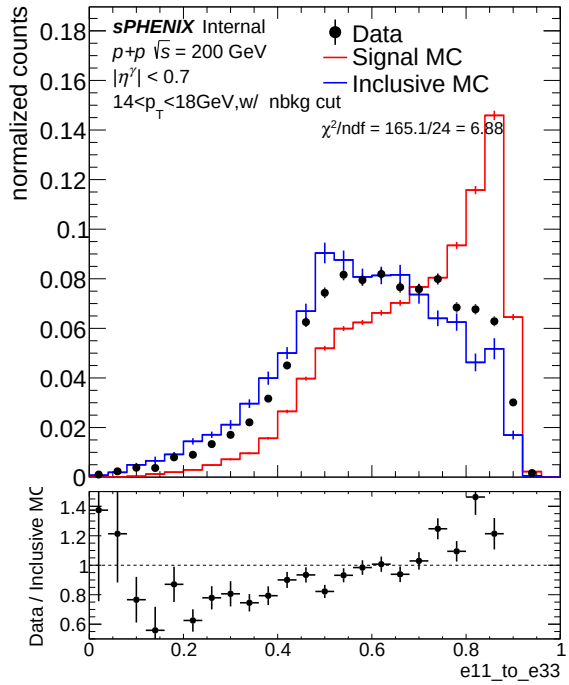
3.3 f_1 and $E_{1\times1}/E_{3\times3}$



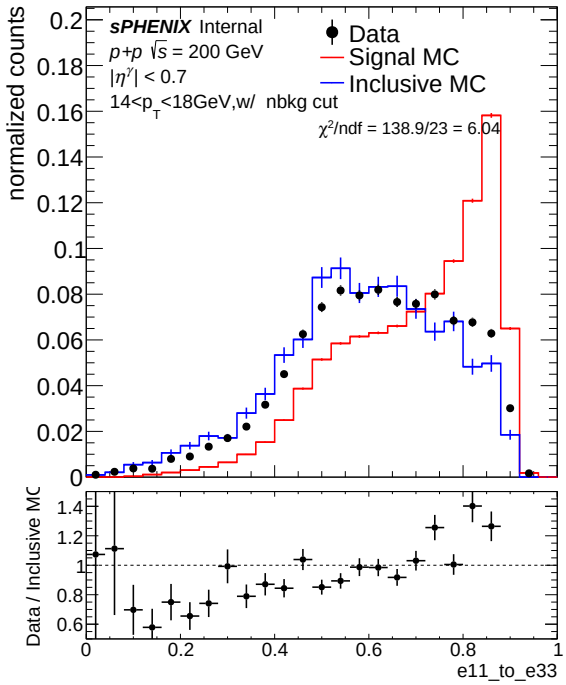
(a) f_1, Nominal MC



(b) f_1, Mixed MC (7.2% DI)



(c) $E_{1\times1}/E_{3\times3}$, Nominal MC

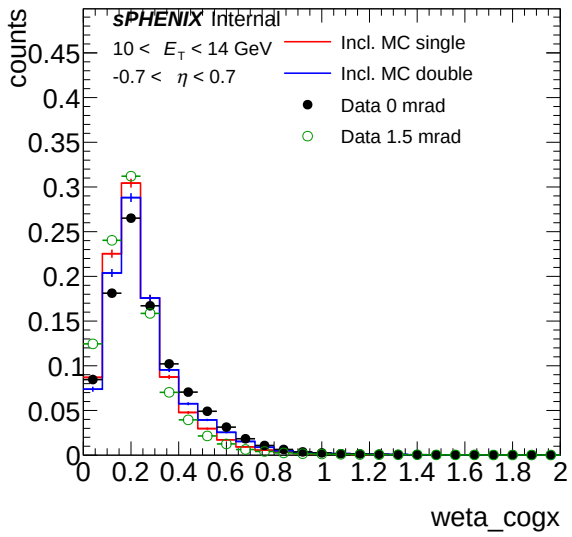


(d) $E_{1\times1}/E_{3\times3}$, Mixed MC (7.2% DI)

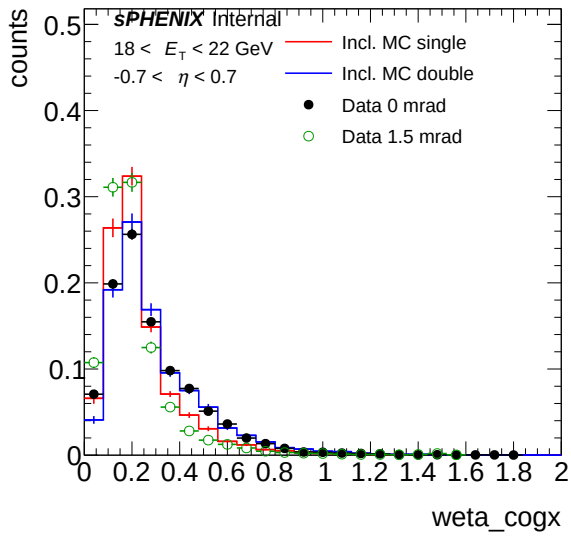
Figure 14: f_1 (top) and $E_{1\times1}/E_{3\times3}$ (bottom) at $14 < E_T < 18\,\text{GeV}$ (pt1), 1.5 mrad. f_1: $\chi^2/\text{ndf} = 4.20 \rightarrow 2.56$ (1.6×). $E_{1\times1}/E_{3\times3}$: $\chi^2/\text{ndf} = 6.88 \rightarrow 6.04$ (1.1×).

4 Single vs Double Interaction Overlay

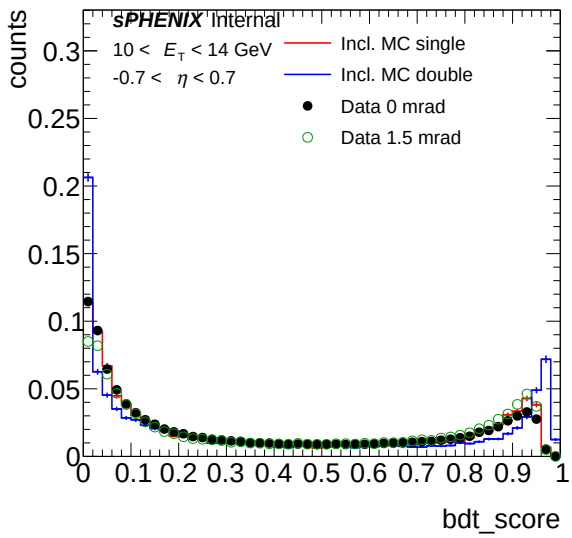
Figure 15 overlays single-interaction MC (red) and double-interaction MC (blue) with data from both crossing-angle periods: 0 mrad (filled black circles) and 1.5 mrad (open green circles), all area-normalized. Data at 0 mrad shows larger deviation from single MC, consistent with the higher pileup fraction, while data at 1.5 mrad tracks closer to single MC.



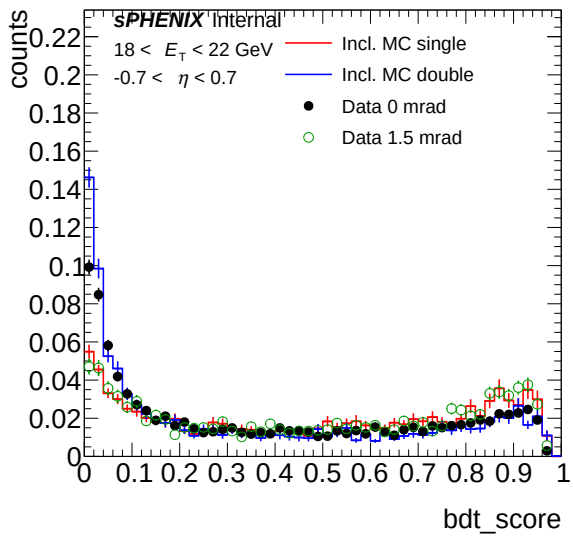
(a) w_η^{CoG}, 10–14 GeV



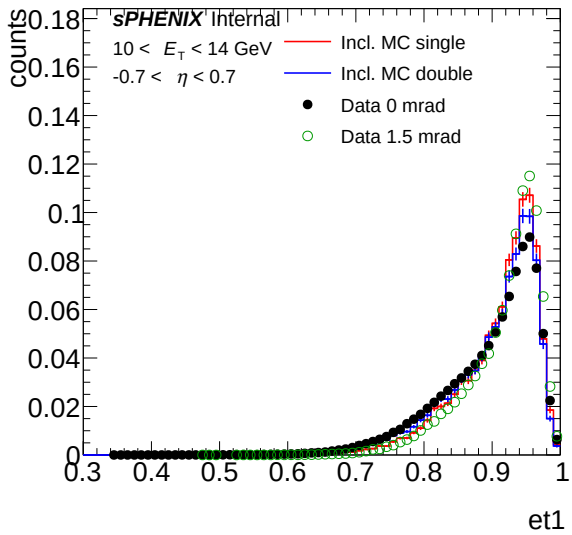
(b) w_η^{CoG}, 18–22 GeV



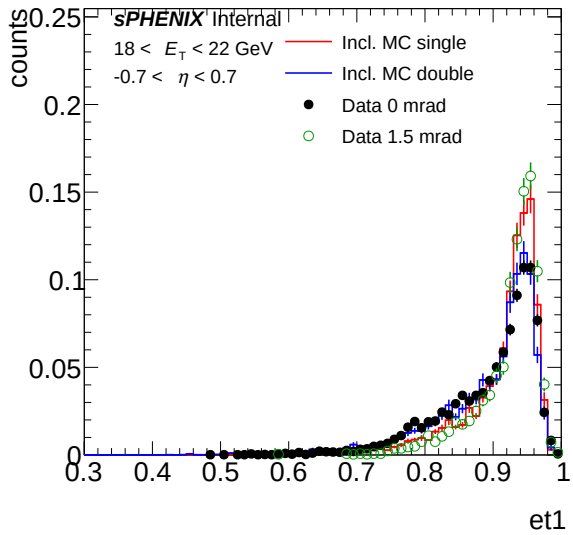
(c) BDT score, 10–14 GeV



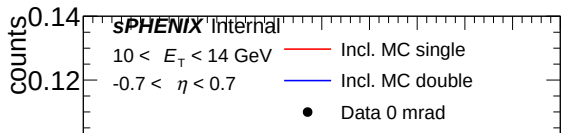
(d) BDT score, 18–22 GeV

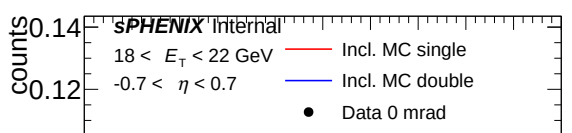


(e) f_1, 10–14 GeV



(f) f_1, 18–22 GeV





5 χ^2/ndf Summary

Table 1 compiles the Data–Inclusive MC χ^2/ndf values for all variables, E_T bins, and crossing angles. Entries where the mixed MC *degrades* agreement relative to nominal are highlighted in red. Improvement factors are computed as Nominal/Mixed when the mixed MC improves, and Mixed/Nominal (prefixed with a minus sign) when it degrades.

Table 1: Data–Inclusive MC χ^2/ndf at cut1 for nominal (single-only) and mixed (double-blended) MC. Lower is better.

		0 mrad (18.7% DI)			**1.5 mrad (7.2% DI)**		
Variable	E_T [GeV]	Nom.	Mixed	Factor	Nom.	Mixed	Factor
w_η^{CoG}	10–14 (pt0)	47.37	29.08	1.6×	9.72	42.34	−4.4×
w_η^{CoG}	14–18 (pt1)	17.63	2.99	5.9×	6.67	4.85	1.4×
w_η^{CoG}	22–28 (pt3)	16.88	0.75	22.5×	21.42	1.31	16.3×
BDT score	10–14 (pt0)	6.85	20.41	−3.0×	5.99	33.18	−5.5×
BDT score	14–18 (pt1)	12.18	1.63	7.5×	1.16	2.31	−2.0×
BDT score	22–28 (pt3)	9.62	1.74	5.5×	1.91	1.13	1.7×
f_1	14–18 (pt1)	9.33	1.93	4.8×	4.20	2.56	1.6×
$E_{1\times1}/E_{3\times3}$	14–18 (pt1)	4.51	4.14	1.1×	6.88	6.04	1.1×

6 Discussion

6.1 Universal pattern: E_T-dependent crossover

A consistent pattern emerges across *both* crossing angles: the double-interaction blending **degrades** data–MC agreement for the BDT score at low E_T (pt0, 10–14 GeV) while **improving** it at higher E_T. This is now confirmed for both 0 mrad (3.0× BDT degradation at pt0) and 1.5 mrad (5.5× BDT degradation at pt0). For w_η^{CoG}, the crossover occurs between pt0 and pt1: at 0 mrad, the pt0 w_η^{CoG} still improves (1.6×), but at 1.5 mrad it degrades (4.4×), indicating that the crossover pileup fraction for w_η^{CoG} lies between 7% and 19% at low E_T.

6.2 Crossing-angle comparison

The improvement factors scale with the pileup fraction: the 18.7% admixture at 0 mrad yields larger improvements (up to 22.5× for w_η^{CoG} at pt3) than the 7.2% admixture at 1.5 mrad (up to 16.3×). Conversely, the 1.5 mrad nominals are already much better than their 0 mrad counterparts (e.g., BDT pt1: 1.16 vs. 12.18), confirming that lower pileup naturally reduces the data–MC discrepancy.

6.3 Low-E_T degradation mechanism

The low-E_T degradation affects the BDT score most severely because the BDT amplifies subtle shower shape distortions through its nonlinear decision function. Contributing factors include:

1. **Shared MC samples**: the same `photon10_double` and `jet12_double` samples are used for both crossing angles; beam conditions in the MC may better match one period than the other.

2. **Low-E_T sensitivity**: clusters near the 8–14 GeV threshold are most sensitive to vertex shifts and energy redistribution.

3. **Small-fraction noise**: at 7.2% mixing, any mismatch between the double MC and real pileup gets amplified relative to the small physical correction.

7 Conclusions and Recommendations

1. **0 mrad period**: the 18.7% double-interaction blending substantially improves data–MC agreement for pileup-sensitive variables (w_η^{CoG}, BDT, f_1) at $E_T \geq 14\,\mathrm{GeV}$, with improvement factors up to 22.5×. At pt0 (10–14 GeV), the BDT score degrades by 3.0× while w_η^{CoG} still improves modestly.

2. **1.5 mrad period**: the 7.2% blending helps at high E_T ($\geq 18\,\mathrm{GeV}$) but degrades both w_η^{CoG} and BDT at low E_T. The nominal agreement is already acceptable ($\chi^2/\mathrm{ndf} \lesssim 10$ for most variables).

3. **Recommended baseline for cross-section measurement**:

   - **0 mrad,** $E_T \geq 14\,\mathbf{GeV}$: use mixed MC (18.7% DI)
   - **0 mrad,** $E_T < 14\,\mathbf{GeV}$: use mixed MC for w_η^{CoG}; use nominal MC for BDT-sensitive quantities; assign nominal–mixed difference as systematic
   - **1.5 mrad,** $E_T \geq 18\,\mathbf{GeV}$: use mixed MC (7.2% DI)
   - **1.5 mrad,** $E_T < 18\,\mathbf{GeV}$: use nominal MC; assign nominal–mixed difference as systematic

4. **Alternative**: use nominal MC for the full 1.5 mrad period and apply the double-interaction correction to 0 mrad data only.